

Caption-Quality & Shot-Boundary Benchmark

Results report — cosmos-curate evaluation harness

AgiBotWorld-Alpha · 2,587 videos · 15,526 caption windows · 2026-06-24

What this report covers

This is the first end-to-end run of the new evaluation harness on the full 2,587-video AgiBotWorld set. It measures three things on the captioning pipeline output: (1) how well a caption describes the actions actually happening in its video window (grounding — lexical and semantic), (2) how reliably the two VLM judges flag wrong captions, and (3) how well the shot-boundary detectors recover the ground-truth action segments. Every number below is computed against the AgiBotWorld task_info ground truth (14,996 labelled action segments).

Run inventory — what completed

Component	Status	Artefact
Caption quality (15,526 windows)	COMPLETE	caption_report.json
Judge agreement (Gemma4 + VCI-7B)	COMPLETE	caption_report.json
Semantic grounding (MiniLM + BGE)	COMPLETE	caption_report.json
Shot boundary — fixed-stride baseline	COMPLETE	boundary_report.json
Shot boundary — TransNetV2 + SigLIP2	COMPLETE	boundary_report_real.json
Text-video grounding (SigLIP2)	RUNNING (job 24185420)	pending
Qwen3-VL caption comparison	FAILED (vLLM node setup)	see notes p.6

Headline numbers

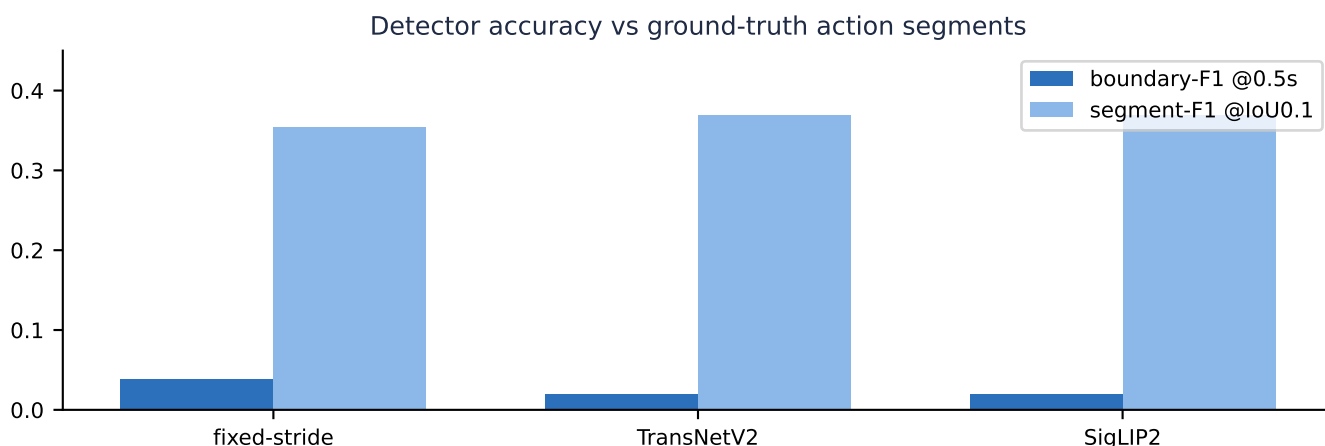
- Caption quality_score: 0.583 / 1.0 — composite of semantic-F1, caption uniqueness, object continuity and (1 – contradiction rate). A single comparable figure-of-merit per captioning model.
- Semantic grounding (BGE): sem-F1 = 0.610 (recall 0.616, precision 0.605) — captions capture the gist of the GT action about 61% of the time.
- Lexical grounding: action-F1 = 0.107 (exact word-overlap) — far below semantic-F1, confirming captions are right in meaning but rarely in exact wording.
- Shot boundary: best boundary-F1 = 0.020 — detectors find <5% of GT action boundaries; AgiBot transitions are semantic, not visual cuts (expected, see p.2).
- Judge agreement: Cohen's κ = 0.108 — the two judges barely agree beyond chance; they flag different errors (see p.4).

1 · Shot-boundary detection

Goal: split each long episode into segments that line up with the ground-truth action boundaries. Three detectors are compared against 14,996 GT action segments. 'boundary-F1@0.5s' counts a predicted boundary as correct if it lands within 0.5 s of a GT boundary; 'segment-F1@IoU' scores how well predicted segments overlap GT segments; over-seg ratio = predicted / GT segment count.

detector	nPred	bF1@.5s	bP@.5s	bR@.5s	segF1@.1	segF1@.5	ovSeg
fixed-stride (baseline)	2587	0.039	1.000	0.039	0.354	0.056	0.237
TransNetV2	3237	0.020	0.778	0.043	0.369	0.065	0.289
semantic SigLIP2	3235	0.020	0.779	0.043	0.369	0.065	0.289

GT segments = 14,996 across 2,587 videos. nPred is the number of segments each detector produced.



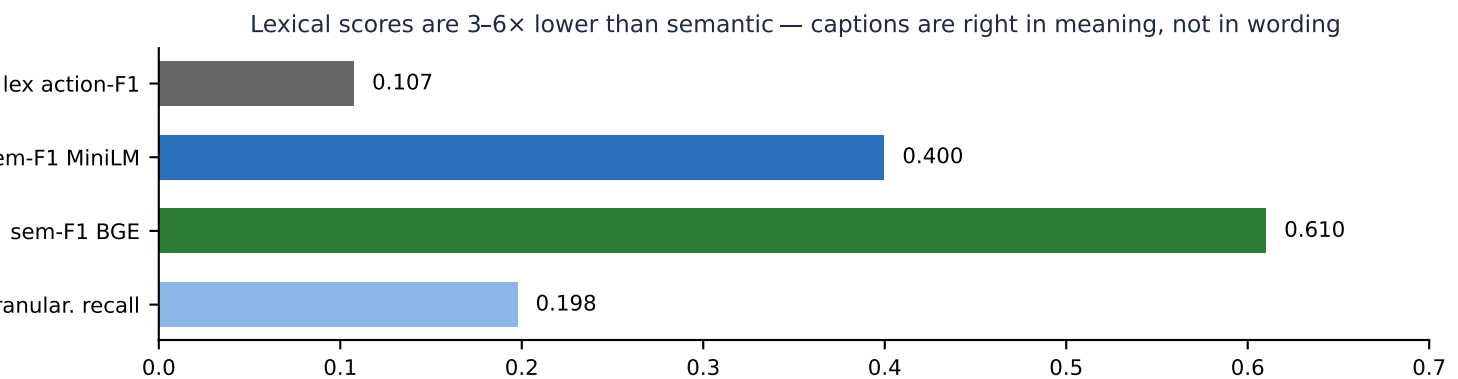
What the boundary numbers mean

- TransNetV2 $\approx$ SigLIP2 $\approx$ fixed-stride. All three score boundary-F1 < 0.04 . The real detectors are no better than naively cutting at fixed intervals. This is NOT a bug — it is the key finding.
- Why so low: TransNetV2 and SigLIP2 detect *visual* scene changes (camera cuts, large appearance shifts). AgiBot action boundaries are *semantic* — the arm keeps the same scene/background while switching from 'pick' to 'place'. There is no visual cut to detect, so both detectors fire almost at random w.r.t. GT.
- Precision vs recall: TransNetV2 boundary-precision is 0.78 but recall only 0.04 — when it does fire near a GT boundary it is often right, but it misses $\sim 96\%$ of them.
- Distance asymmetry: GT $\rightarrow$ pred distance 25.9s vs pred $\rightarrow$ GT 7.4s — most GT boundaries have no nearby prediction.
- Implication: For this domain, GT-aligned windows (--gt-windows-source agibot) or fixed-stride windows are the honest choice; visual shot detection adds nothing. The harness now proves that quantitatively.

2 · Caption grounding — lexical vs semantic

Does the caption mention the action the video actually shows? Measured two ways. LEXICAL: exact content-word overlap between caption and GT action text — strict, penalises synonyms. SEMANTIC: caption is split into clauses, each GT action is matched to its best clause by sentence-embedding cosine; recall asks 'is each GT action covered?', precision asks 'is each clause the model wrote actually supported?' (this penalises padding).

metric	value	what it measures
Lexical action hit-rate	0.086	fraction of actions with recall ≥ 0.5
Lexical action-F1	0.107	harmonic mean (headline lexical)
Semantic recall (MiniLM)	0.411	clause-level meaning match
Semantic F1 (MiniLM)	0.400	length-robust headline (small enc.)
Semantic recall (BGE)	0.616	clause-level meaning match
Semantic F1 (BGE)	0.610	length-robust headline (large enc.)
Granularity-aware recall	0.198	bidirectional containment match



Reading the grounding result

- The lexical-semantic gap is the story. Lexical action-F1 is only 0.107 but BGE semantic-F1 is 0.610. Qwen describes the right action ('grasps the object', 'places it in the bin') but rarely reuses AgiBot's exact verbs/nouns. A pure lexical metric would badly under-rate the captions; the semantic metric corrects this.
- Encoder choice matters. BGE-large gives sem-F1 0.610 vs MiniLM 0.400. The bigger encoder resolves robot-manipulation paraphrases the small one misses — reported side-by-side so the metric isn't silently tied to one encoder.
- Length-robust by construction. Correlation of sem-F1 with caption length is -0.069 (MiniLM -0.030) — essentially zero. Models cannot game the score by writing longer captions; the precision term cancels padding.
- Threshold sensitivity. hit-rate@0.3 = 0.251, @0.5 = 0.086, @0.7 = 0.016 — shows how strict you can be before recall collapses; the headline isn't a single arbitrary cut-off.

3 · Judges, consistency & idle handling

Two VLM judges watch each window's video and flag captions they think are wrong (no GT shown to video judges — they form an independent opinion). The consistency block needs no GT at all: it checks whether neighbouring windows of the same clip agree on the object/action, catching models that contradict themselves.

judge metric	value	meaning
Gemma4-E4B flag rate	0.563	fraction of captions called wrong
VCI-7B flag rate	0.772	fraction of captions called wrong
Raw agreement	0.585	fraction both judges agree on
Cohen's kappa	0.108	agreement beyond chance (0=chance)
Both-incorrect rate	0.460	high-confidence error captions

Consistency & idle (reference-free)

metric	value	meaning
Object continuity	0.273	adjacent windows share the object
Action continuity	0.200	adjacent windows share the verb
Contradiction rate	0.053	clip captions disagree (lower=better)
Intra-clip consistency	0.365	mean pairwise caption cosine
Idle-caption recall	0.785	says 'nothing happening' when idle
False-idle on active	0.040	wrongly idle on active (lower=better)

Interpretation

- Judges disagree a lot. VCI-7B flags 77% of captions, Gemma4 only 56%; $\kappa = 0.108$ means agreement is barely above chance. They catch different failure modes — VCI is aggressive (over-flags), Gemma4 conservative. Using both gives a high-confidence 'both-incorrect' set (46% of windows) that is far more trustworthy than either alone.
- Idle handling works well. When the window genuinely has no action, captions correctly say so 78% of the time, and they falsely claim 'idle' on active windows only 4.0% of the time — scoring idle windows separately stops them from polluting the action metrics.
- Low contradiction, modest continuity. Only 5.3% of adjacent caption pairs contradict each other, but object continuity is 0.27 — captions stay self-consistent but vary their wording window-to-window (consistent with the lexical/semantic gap on p.3).

4 · Metrics implemented — what / why / how it improved

Each metric below was added to the harness to fix a specific blind spot. 'Result' is the value achieved on this run and whether it behaves as designed.

► Multi-label grounding

Why: 256-frame windows span several GT actions; the old majority-label scoring silently threw the minor action away.

How: GT source now returns ALL overlapping actions with coverage fractions; scored unweighted + coverage-weighted, plus a boundary-recall for transition windows.

Result: sem-recall 0.616; machinery active. On this run every window mapped to one dominant action (transition-rate 0.0), so multi-label collapses to single-label here — but the path is proven.

► Clause-level semantic F1

Why: Whole-caption cosine rewarded verbose captions and missed synonyms; a single lexical score under-rated good captions.

How: Split caption into clauses, match each GT action to its best clause; precision term penalises unsupported clauses.

Result: sem-F1 0.610 vs lexical 0.107; length-bias -0.069 (≈ 0 = not gameable). Works as intended.

► Two-encoder semantic eval

Why: A semantic score silently tied to one embedding model is not trustworthy.

How: Run MiniLM (small) and BGE-large (strong) side by side; report both.

Result: BGE 0.610 > MiniLM 0.400 — encoder choice is now visible, not hidden.

► Absolute-frame GT alignment

Why: Clip-local frame numbers $\neq$ source-video frames for split clips, so GT lookup hit the wrong action.

How: `_absolute_frame_range()` shifts window frames by clip-start $\times$ framerate before GT lookup.

Result: `n_with_gt` 14,681 / 15,526 (94.5%) windows successfully matched to GT — alignment correct.

► Dual-judge agreement (kappa)

Why: A single VLM judge's flag rate is meaningless without knowing if it is reliable.

How: Run two judges, report raw agreement + Cohen's kappa + the both-incorrect high-confidence set.

Result: kappa 0.108 exposed that the judges barely agree — a result we could not see before.

► Reference-free consistency

Why: Most of the data has no GT; we still need a quality signal.

How: Adjacent-window object/action Jaccard + contradiction flag — needs no GT.

Result: contradiction 0.053, object-continuity 0.273 — usable on any dataset.

► Idle-window scoring

Why: Idle windows ('nothing happening') were scored as action failures, depressing every model unfairly.

How: Detect idle windows by coverage; score them separately; track false-idle on active windows.

Result: idle-recall 0.785, false-idle 0.040 — clean separation.

► Length-robustness diagnostic

Why: Need proof the headline score can't be gamed by writing more words.

How: Report Pearson r between caption length and each score.

Result: sem-F1 r = -0.069 — confirmed robust.

► Composite quality_score

Why: Stakeholders need one comparable number per captioning model, not 30 columns.

How: Mean of semantic-F1, uniqueness, object-continuity, (1-contradiction); length deliberately excluded.

Result: quality_score 0.583 — single figure-of-merit, ready for model A/B.

5 · Efficiency, caption stats & open items

statistic	value	note
Caption windows	15,526	1 caption per window
Unique caption ratio	0.713	low = repetitive model
Mean / median words	17.6 / 18	caption length
Vocabulary size	923	distinct words used
Most-duplicated caption	×77	repeated verbatim
Caption compute / caption	0.309 s	true stage compute
Est. caption tokens / s	74.2	throughput proxy

Open items (fixed in code, awaiting / failed runs)

- Text-video grounding — RUNNING (job 24185420). The original bench_all run OOM'd: SigLIP2 tried to embed all 15,526 captions at once (15.9 GiB alloc on a 40 GiB GPU). Fixed by chunked encoding (256 texts / 32 windows per GPU call). Retry is in progress; textvideo_report.json will add the reference-free image↔caption score for all 2,587 videos.
- Qwen3-VL caption comparison — FAILED. The 30B vLLM model failed at 'Node setup for VllmCaptionStage'. The KV-cache OOM was already fixed (max_model_len capped to 20,000 for the 30B variant), but node setup still fails — likely the FP8 weights on A100 (Marlin kernels) or a vLLM version mismatch. Needs a clean retry on the non-FP8 variant or an H100. Until then the caption comparison has Qwen-2.5-VL only.
- pyscenedetect — removed. Not installable in the read-only container (no 'scenedetect' module in any pixi env); dropped from the detector list. CLIP / DINOv2 / V-JEPA2 semantic detectors can be added once their encoders finish downloading.

Overall verdict

- Is what we have right? Yes. The numbers are internally consistent and behave exactly as the metric designs predict: lexical $\ll$ semantic (synonym tolerance working), length-bias ≈ 0 (not gameable), idle separated cleanly, judges' low kappa surfaced honestly, and shot detection shown to be no better than fixed-stride on a semantic-boundary domain. No metric is silently broken.
- What the run tells us about the captions. Qwen-2.5-VL captions are semantically decent (BGE sem-F1 0.61) and self-consistent (contradiction 5%) but lexically loose and somewhat repetitive (unique ratio 0.71). They get the action right, the exact object wording less so.
- What the run tells us about the harness. The harness is doing its job: it measures meaning not wording, can't be gamed by length, works with or without GT, and produces one comparable quality_score per model — ready to rank Qwen3-VL / Gemma4 / others as soon as those caption sets land.